

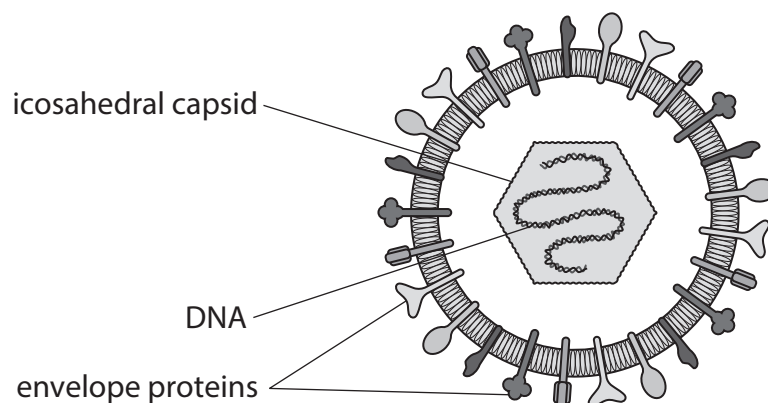
5 Herpes simplex is a virus that causes sores around the nose and mouth.

A modified form of this virus has been developed to infect and kill cancer cells in humans.

(a) Explain how a cell can become a cancer cell.

(2)

(b) The diagram shows a herpes simplex virus.



(i) Which of the following statements are true?

(1)

- 1 Ebola virus has the same type of genetic material as herpes simplex virus
- 2 Both Ebola virus and human immunodeficiency virus (HIV) have envelope proteins
- 3 HIV has an icosahedral capsid

- ☐ **A** Statements 1 and 2 only
- ☐ **B** Statements 1 and 3 only
- ☐ **C** Statements 2 and 3 only
- ☐ **D** Statements 1, 2 and 3

(ii) State the function of the envelope proteins.

(1)

(iii) Explain why the herpes simplex virus has to be modified before it can be used to infect cancer cells.

(2)

(iv) Explain how the infection of a cancer cell with herpes simplex virus can result in the death of this cell.

(2)

(c) Using a modified herpes simplex virus activates the immune system.

Explain why activation of the immune system would result in destruction of the cancer cells.

(4)

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(Total for Question 5 = 12 marks)
